Person	Date	Exam_Period	Coffee_Cups	Sleep_Hours	Study_Hours	Stress_Levels
Ayşe	1.04.2026	0	1	8.0	2	3
Ayşe	2.04.2026	0	2	7.5	3	4
Ayşe	3.04.2026	0	1	7.0	4	4
Ayşe	4.04.2026	1	4	6.0	8	7
Ayşe	5.04.2026	1	5	5.0	10	9
Ayşe	6.04.2026	1	3	5.5	9	8
Ayşe	7.04.2026	1	4	4.5	11	10
Ayşe	8.04.2026	0	1	9.0	1	2
Elif	1.04.2026	0	0	8.0	1	2
Elif	2.04.2026	0	1	7.5	2	2
Elif	3.04.2026	0	1	8.0	2	3
Elif	4.04.2026	0	2	7.0	3	4
Elif	5.04.2026	1	3	6.0	7	6
Elif	6.04.2026	1	4	5.0	9	8
Elif	7.04.2026	1	5	5.5	10	9
Elif	8.04.2026	1	4	4.0	12	10
Berfin	1.04.2026	1	3	6.0	8	7
Berfin	2.04.2026	0	1	9.0	0	1
Berfin	3.04.2026	0	2	7.0	4	5
Berfin	4.04.2026	0	2	7.5	3	4
Berfin	5.04.2026	1	4	5.0	8	8
Berfin	6.04.2026	1	5	4.0	10	9
Berfin	7.04.2026	1	6	4.5	11	10
Berfin	8.04.2026	1	4	5.5	9	7
Emre	1.04.2026	1	3	6.0	7	6
Emre	2.04.2026	0	1	8.0	2	3
Emre	3.04.2026	0	0	8.5	1	2
Emre	4.04.2026	0	1	7.0	3	4
Emre	5.04.2026	0	2	7.0	4	5
Emre	6.04.2026	0	1	8.0	2	3
Emre	7.04.2026	1	3	6.0	6	7
Emre	8.04.2026	1	4	5.5	8	8
İlayda	1.04.2026	1	5	4.0	11	10
İlayda	2.04.2026	1	4	5.0	9	9
İlayda	3.04.2026	0	2	7.5	3	4
İlayda	4.04.2026	0	1	8.0	1	2
İlayda	5.04.2026	0	0	8.5	2	2
İlayda	6.04.2026	0	1	8.0	3	3
İlayda	7.04.2026	0	1	7.5	4	4
İlayda	8.04.2026	1	2	6.0	7	6
Zeynep	1.04.2026	1	3	5.5	9	8
Zeynep	2.04.2026	1	4	5.0	10	9
Zeynep	3.04.2026	1	5	4.5	12	10
Zeynep	4.04.2026	1	4	5.0	11	9
Zeynep	5.04.2026	0	1	8.0	2	4
Zeynep	6.04.2026	0	0	9.0	1	2
Zeynep	7.04.2026	0	2	7.0	4	5
Zeynep	8.04.2026	0	2	7.0	5	6
Sude	1.04.2026	1	4	5.5	8	8
Sude	2.04.2026	1	5	5.0	10	9
Sude	3.04.2026	1	5	4.5	11	10
Sude	4.04.2026	1	4	6.0	9	8
Sude	5.04.2026	1	3	6.5	7	7
Sude	6.04.2026	0	1	8.0	3	4
Sude	7.04.2026	0	1	8.5	2	3
Sude	8.04.2026	0	1	8.0	2	2
Alp	1.04.2026	0	1	7.5	3	3
Alp	2.04.2026	0	2	7.0	4	4
Alp	3.04.2026	1	4	5.0	9	8
Alp	4.04.2026	1	5	4.0	11	10
Alp	5.04.2026	1	4	5.5	10	9
Alp	6.04.2026	1	3	6.0	8	7
Alp	7.04.2026	0	1	8.0	2	3
Alp	8.04.2026	0	1	8.5	1	2

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind

df = pd.read_csv("coffee_data.csv")

df["Sleep_Hours"] = df["Sleep_Hours"].astype(float)

print("\n--- DATA OVERVIEW ---")
print(df.head())
print(df.describe())

print("\n--- AVERAGE COFFEE ---")
print(df.groupby("Exam_Period")["Coffee_Cups"].mean())

plt.figure()
plt.hist(df["Coffee_Cups"])
plt.title("Coffee Distribution")
plt.show()

plt.figure()
sns.boxplot(x="Exam_Period", y="Coffee_Cups", data=df)
plt.title("Exam vs Coffee")
plt.show()


exam = df[df["Exam_Period"] == 1]["Coffee_Cups"]
normal = df[df["Exam_Period"] == 0]["Coffee_Cups"]

t_stat, p_value = ttest_ind(exam, normal)

print("\n--- T-TEST ---")
print("T-stat:", t_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Result: Coffee consumption is significantly higher during
exams")
else:
    print("Result: No significant difference")
```

```
--- T-TEST ---
T-stat: 14.88265866383516
P-value: 3.899500882751304e-22
Result: Coffee consumption is significantly higher during exams
```

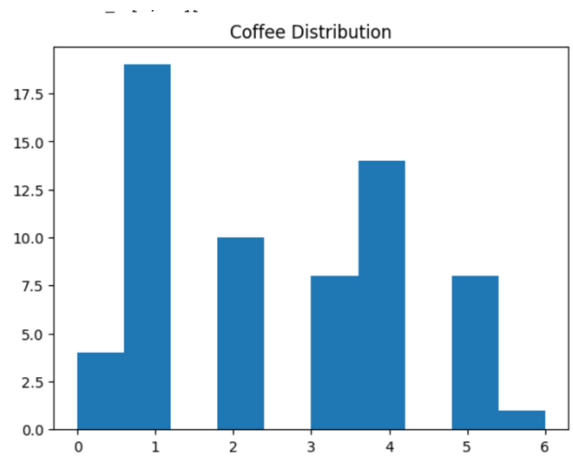
H0 (Null Hypothesis): There is no difference in coffee consumption between exam and non-exam periods.

H1 (Alternative Hypothesis): Coffee consumption is higher during exam periods.

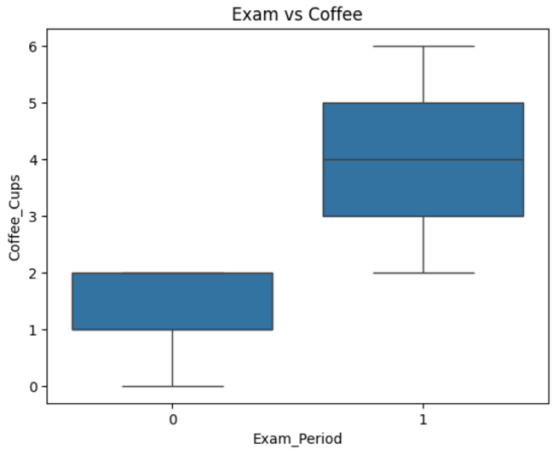
A two-sample t-test was conducted to compare the mean coffee consumption between exam and non-exam days.

- **T-statistic:** 14.88
- **P-value:** 3.89e-22

The results of the t-test show a very small p-value (p ≈ 3.89e-22), which is significantly lower than the significance level of 0.05. So, the null hypothesis is rejected. This indicates that there is a statistically significant difference in coffee consumption between exam and non-exam periods. Furthermore, the positive t-statistic suggests that coffee consumption is significantly higher during exam periods.



The histogram shows the distribution of daily coffee consumption across all observations. Most values are concentrated between 1 and 4 cups per day, while higher values (5–6 cups) occur less frequently. This indicates that moderate coffee consumption is common, but extreme consumption tends to happen only occasionally, likely during exam periods.



The boxplot compares coffee consumption between exam and non-exam periods. It clearly shows that the median coffee consumption is significantly higher during exam periods. Additionally, the spread of values is wider during exams, indicating more variability in consumption. This suggests that students tend to consume more coffee and show more extreme behaviors during exam periods.